

## Library Book Usage & Reading Pattern Analysis

A data analysis and visualization mini-project that studies how students borrow and use library books — identifying the most popular books, busiest departments, seasonal borrowing trends, and average reading duration. Built with **Python, Flask, MySQL, and HTML/CSS**.

- **Domain:** Data Analysis & Visualization
  - **Technology:** Python, Flask, MySQL, HTML/CSS
  - **Type:** Individual Mini Project — AIDS, Anna University
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### Problem Statement

Libraries in educational institutions maintain large collections of books, but lack proper tools to track and analyze borrowing patterns. There's no clear understanding of which books are frequently used, which departments are most active, and during what periods library usage is highest. This project uses real borrowing data to generate actionable insight for

library management decisions — what to restock, what to remove, and when to extend library hours.

## Module Breakdown

| Module          | Responsibility                                           | Tech Used                                              |
|-----------------|----------------------------------------------------------|--------------------------------------------------------|
| Database        | Stores admin, book, student, and borrow-record data      | MySQL<br>( database.sql )                              |
| Data Analysis   | Loads borrow data, computes stats, generates charts      | Python, Pandas, Matplotlib, Seaborn<br>( analysis.py ) |
| Web Application | Login-protected dashboard, analysis views, records table | Flask ( app.py )                                       |
| Frontend        | Displays dashboard, charts, and tables                   | HTML/CSS<br>( templates/ )                             |

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## Database Schema

### Table: admin

| Column   | Type         | Notes                       |
|----------|--------------|-----------------------------|
| admin_id | INT          | Primary Key, auto-increment |
| username | VARCHAR(50)  | Unique                      |
| password | VARCHAR(255) |                             |
| email    | VARCHAR(100) |                             |
| role     | VARCHAR(20)  | Default: 'admin'            |

### Table: books

| Column           | Type         | Notes       |
|------------------|--------------|-------------|
| book_id          | VARCHAR(10)  | Primary Key |
| title            | VARCHAR(200) |             |
| author           | VARCHAR(100) |             |
| genre            | VARCHAR(50)  |             |
| total_copies     | INT          | Default: 1  |
| available_copies | INT          | Default: 1  |

### Table: students

| Column     | Type         | Notes       |
|------------|--------------|-------------|
| student_id | VARCHAR(10)  | Primary Key |
| name       | VARCHAR(100) |             |
| department | VARCHAR(20)  |             |
| year       | INT          |             |

| email | VARCHAR(100) | |  
| roll\_number | VARCHAR(20) | |

**Table: borrow\_records**

| Column | Type | Notes |  
|--|--|--|  
| record\_id | INT | Primary Key, auto-increment |  
| student\_id | VARCHAR(10) | Foreign Key → students |  
| book\_id | VARCHAR(10) | Foreign Key → books |  
| borrow\_date | DATE | |  
| return\_date | DATE | |  
| due\_date | DATE | |  
| status | VARCHAR(20) | Default: 'Borrowed' |

**Table: reports**

| Column | Type | Notes |  
|--|--|--|  
| report\_id | INT | Primary Key, auto-increment |  
| generated\_by | INT | Foreign Key → admin |  
| report\_type | VARCHAR(50) | |  
| generated\_date | DATE | Default: current date |  
| description | TEXT | |

The database ships pre-loaded with a default admin account and 15 sample books across Technical, Science, Self Help, History, and Fiction genres.

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 **How It Works**

## 1. Data Storage

`database.sql` sets up the MySQL database and 5 tables, and seeds sample admin/book data.

## 2. Analysis ( `analysis.py` )

- Loads `data/library_data.csv`
- Parses borrow/return dates, calculates holding duration ( `hold_days` ) and borrow month
- Computes summary stats: most borrowed book, top department, peak month, top genre, average hold time
- Generates 6 charts and saves them as PNGs in `static/charts/` :
  -  Top 10 most borrowed books (bar chart)
  -  Genre-wise distribution (pie chart)
  -  Monthly borrowing trend (line chart)
  -  Department-wise activity (bar chart)
  -  Department vs. month heatmap
  -  Year-wise reading activity (bar chart)

## 3. Web App ( `app.py` )

- `/login` — admin login (session-based)
- `/` — dashboard: regenerates charts, shows summary stats
- `/analysis` — top 5 books and department breakdown
- `/records` — table of borrow records

- `/logout` – clears session

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## ✓ Sample Results

| Metric                 | Example Result                                 |
|------------------------|------------------------------------------------|
| Most borrowed book     | <i>Data Structures</i> by Narasimha Karumanchi |
| Most active department | CSE / AIDS                                     |
| Peak reading month     | November (exam season)                         |
| Least read genre       | Fiction                                        |
| Avg. book holding time | 7 days                                         |

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## 📁 Project Structure

```
Library-Book-Usage-and-Reading-Pattern-Analysis/  
|  
├── data/  
|   └── library_data.csv  
└── app.py                # Flask app:
```

```
login, dashboard, routes
├── analysis.py          # Data analysis
+ chart generation
├── database.sql         # MySQL schema
+ sample data
├── requirements.txt
├── static/
│   └── charts/          # generated
chart images
├── templates/
│   ├── login.html
│   ├── index.html
│   ├── analysis.html
│   └── records.html
└── README.md
```

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## Setup Instructions

### 1. Clone the repository

```
git clone
https://github.com/sunderpradeep14-
blip/Library-Book-Usage-and-Reading-
Pattern-Analysis.git
cd Library-Book-Usage-and-Reading-
Pattern-Analysis
```

### 2. Create a virtual environment

```
python -m venv venv
source venv/bin/activate    # On
Windows: venv\Scripts\activate
```

### 3. Install dependencies

```
pip install -r requirements.txt
```

### 4. Set up the MySQL database

```
mysql -u root -p < database.sql
```

### 5. Run the Flask app

```
python app.py
```

6. Open `http://127.0.0.1:5000` in your browser and log in with:

- **Username:** `admin`
- **Password:** `admin123`

⚠ These are demo credentials for local/academic use only — not intended for production deployment.

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 **Tech Stack**



- **Backend:** Python, Flask
  - **Database:** MySQL
  - **Frontend:** HTML, CSS
  - **Data Analysis:** Pandas
  - **Visualization:** Matplotlib, Seaborn
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## Why This Project Stands Out

- 🏠 Directly relevant to everyday college life
  - 🔒 Includes a real login-protected admin dashboard, not just static charts
  - 📊 6 distinct visualizations covering books, genres, time trends, and departments
  - 🧩 Clean 5-table relational schema with proper foreign keys
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## Future Enhancements

- Student-facing login to view personal borrowing history
  - Predictive model for book demand forecasting
  - Overdue book alert/notification system
  - Move admin credentials to environment variables for better security
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## **License**

This project is open-source and available for academic and educational use.